Author's Note

No Single Tool Runs the Pipeline

Preeta Chatterjee | CSYE 7270: Building Virtual Environments | Northeastern University

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Design Choices

Why this topic

This topic chose itself. By the time the midterm arrived, I had already run five distinct AI tool evaluations as part of building Hallway Hunters — floor plan generation, zombie character reference art, walk cycle animation, Niagara system configuration, and blood splatter material design. I had not framed any of them as evaluations while doing them. I was just trying to get assets made. What the midterm forced me to do was look back at those decisions and articulate what had actually happened: why I switched tools, what each tool could and could not do, and where the AI pipeline ended and human judgment began.

The topic also had a natural failure case already built in. The walk cycle flipbook failure was not something I had to construct for the assignment — it happened during production, it was tested across two tools and multiple prompt iterations, and I already knew why it failed before I wrote a word of the essay. That meant the essay could argue from evidence rather than from speculation. That was the reason to choose it.

What I left out and why

I left out the electrical sparks Niagara system, which also involved parameter divergence from Claude's suggestions. The blood splatter case was stronger because the divergence was philosophical — a complete rebuild of the material approach, not just numerical adjustments. The sparks case involved tuning three values: loop duration, sprite size, and light intensity. These were meaningful changes but they were adjustments within a framework Claude had set correctly. The blood splatter case rejected the framework entirely. A grader reading both would find the blood splatter case more convincing as a Human Decision Node because the rejection is harder to dismiss as fine-tuning.

I also left out the ChatGPT versus Claude floor plan comparison as a standalone section and folded it into Decision Point 1 rather than giving it its own heading in the mechanism section. Claude's inability to produce images is architectural and worth one clear statement, but spending more space on it would have implied it was the central argument of the piece. It is not. The central argument is about tool selection across a pipeline, and Claude's image generation limitation is one data point in that argument, not the thesis.

How this essay is a specific instance of the master claim

The course's master claim is: AI is a pipeline decision, not a magic layer. Where you put it, and how you constrain it, determines what your game becomes. This essay demonstrates that claim through five decisions where the pipeline placement of each tool determined the quality of the output. Claude placed in the image generation slot produced unusable floor plan attempts. Claude placed in the structural configuration slot produced correct Niagara setup steps. The same tool, two placements, two completely different outcomes. That is the master claim made concrete. The essay's contribution is showing that the decision is not just about which tool to use — it is about recognizing which slot in the pipeline each tool belongs in, and being willing to exit the pipeline entirely when no tool fits.

Tool Usage

Where Claude generated and where I corrected

Claude was used to generate the first draft of the essay's five-section structure. The Scenario section came back largely usable — the setup of Hallway Hunters and the pipeline challenge was described accurately. The Mechanism section required the most correction. Claude's initial draft described the tools at a surface level — Claude is a language model, ChatGPT generates images — without distinguishing why those architectural differences matter for specific production tasks. I rewrote the mechanism section to focus on input-output structure and what each tool's output space actually contains, rather than what the tools are generically described as doing.

The most significant correction was in Decision Point 4. Claude's first draft of that section described the blood splatter material divergence as parameter tuning — adjusting values to get a better result. That framing was wrong. The divergence was not tuning. It was a rejection of the material philosophy entirely: different blend mode, different shading model, different opacity approach. I rewrote that section to make the philosophical nature of the rejection explicit, because a tuning correction and a philosophy rejection are different categories of human decision and the grader needed to see that distinction.

What Eddy flagged

Eddy was not accessible during the production of this submission. In place of Eddy's audit, I performed a self-directed editorial pass focused on the same criteria the tool is designed to check: passive constructions, unsupported claims, and jargon. I identified and revised passive constructions in the failure case section — phrases that hid the agent of observation — replacing them with direct statements. I also reviewed the conclusion and made a deliberate choice to keep the compressed final line, judging that the essay's argument is about practical discipline and the conclusion should feel like a constraint, not an inspiration.

What Figure Architect produced

Figure Architect was not accessible during the production of this submission. In place of a generated figure prompt, I identified the high-assertion claims in the essay that would benefit from a visual and designed a substitute manually. The claim that required the most visual support was the pipeline summary — which tool was selected for which task and why. Rather than produce a diagram that would imply a clean sequential handoff between tools, I replaced it with the summary table in the notebook's final cell. The real pipeline was not sequential — tool decisions happened in parallel and were revised as production proceeded — and a table communicates that without the misleading linearity of a flow diagram.

Self-Assessment

Scoring against the rubric

Argumentative Rigor (35 points): I would score this 30-32. The essay makes a specific claim, traces the failure case mechanistically from model architecture to observable output, and the failure is triggered and measured in the demo with pixel similarity scores across three sprite sheets. The area where it falls short of full marks is the Exercise section — the exercise is clear and reproducible, but it asks the reader to reproduce the failure rather than showing a modification that triggers a new failure mode. A stronger exercise would have shown a second failure axis, such as what happens when you apply the same prompt to a model not evaluated in the essay.

Technical Implementation (25 points): I would score this 22-24. The demo runs in Colab with real production outputs, the Human Decision Node is present and specific, and the similarity analysis produces measurable numbers that support the written argument. The blood splatter material graph is visible and the node chain is documented. The area where it falls short is that the notebook does not include a runnable implementation of the AI application itself — it documents outputs already produced rather than calling APIs live. This is a legitimate constraint given tool access restrictions, and the documented outputs are real, but a grader looking for a live API call will not find one.

Clarity (20 points): I would score this 18-20. The five-section structure is followed. Every claim has a mechanism. The accessibility scope note appears early and prevents the obvious counterargument from landing. The one risk is that the mechanism section covers four tools in sequence, and a reader skimming might read it as a tool survey rather than an architectural argument. The subheadings are designed to prevent this but cannot fully eliminate the risk.

Relative Quality (20 points): I would place this in the top 25-50%. The Human Decision Node is present, specific, documented in the notebook, and appears on camera in the video — it is not a general statement about AI limitations but a specific rejection of a specific material philosophy with named alternatives. The failure case is mechanistically explained and measurable. What would push this into the unambiguous top 25% is a live API demonstration and a second failure axis in the Exercise. Given the time and access constraints under which this was produced, the submission is as strong as the constraints allow.